🧠 **LINQ Assignments (C#)**

🟢 **Level 1: Basic LINQ Operations**

✅ **Assignment 1: Student Filtering**

Create a Student class with:

Id, Name, Age, Marks

👉 Tasks:

1. Get all students with marks > 75
2. Get students whose age is between 18 and 25
3. Sort students by Marks (descending)
4. Select only Name and Marks

✅ **Assignment 2: Numbers Collection**

Given:

```
List<int> numbers = new List<int> { 5, 10, 15, 20, 25, 30 };
```

👉 Tasks:

1. Get even numbers
2. Get numbers greater than 15
3. Find square of each number
4. Count how many numbers are divisible by 5

✅ **Assignment 3: String Operations**

Given:

```
List<string> names = new List<string> { "Ravi", "Kiran", "Amit", "Raj", "Anil" };
```

👉 Tasks:

1. Get names starting with 'A'
2. Sort names alphabetically
3. Convert all names to uppercase
4. Find names with length > 4

🟡 **Level 2: Intermediate LINQ**

✅ **Assignment 4: Employee Management**

Create:

Id, Name, Department, Salary

👉 Tasks:

1. Get employees from "IT" department
2. Get highest salary employee
3. Get average salary
4. Group employees by Department
5. Count employees in each department

✅ **Assignment 5: Orders and Customers**

Create:

Customer: Id, Name

Order: Id, CustomerId, Amount

👉 Tasks:

1. Join Customers and Orders
2. Get total order amount per customer
3. Get customers with total orders > 5000
4. Get customers who have no orders

✅ Assignment 6: Duplicate Handling

Given:

```
List<int> numbers = new List<int> { 1,2,3,2,4,5,3,6 };
```

👉 Tasks:

1. Remove duplicates
2. Find duplicate values
3. Count occurrence of each number

🔵 Level 3: Advanced LINQ

✅ Assignment 7: Product Inventory

Create:

Id, Name, Category, Price, Stock

👉 Tasks:

1. Get products with stock < 10
2. Get top 3 expensive products
3. Group products by Category
4. Get total stock per category
5. Check if any product is out of stock

✅ Assignment 8: Multi-Level Grouping

Create:

Student: Id, Name, Class, Subject, Marks

👉 Tasks:

1. Group by Class
2. Then group by Subject within each Class

3. Get average marks per subject per class

✅ **Assignment 9: Real-Time Scenario (E-Commerce)**

Create:

Order: Id, CustomerName, OrderDate, TotalAmount

👉 Tasks:

1. Get orders placed in last 30 days
2. Get monthly sales report
3. Get top 5 customers by spending
4. Get highest order per month

🔴 **Level 4: Expert**

✅ **Assignment 10: Custom Sorting**

Sort employees:

1. By Department (ascending)
2. Then by Salary (descending)

✅ **Assignment 15: Complex Dashboard**

Create:

Employee: Id, Name, Department, Salary, JoiningDate

👉 Build dashboard:

1. Total employees
2. Department-wise average salary
3. Recently joined employees (last 6 months)

4. Highest paid employee per department
5. Salary distribution (min, max, avg)